



ANALYSING THE IMPACT OF COVID-19 ON GLOBAL LEARNING AND EDUCATION.

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Introduction

One of the numerous disruptions to our daily lives caused by COVID-19 was in the global educational sector. Students across the globe experienced learning losses due to the COVID-19-related school closures, with some demographic groups being impacted more severely than others (Reuge et al., 2021). Using our dataset, which contains data from 41 countries across 24 variables between March 2020 and March 2022, this report seeks to answer the following research question: How has the COVID-19 pandemic affected learning loss among students, and what factors have contributed to this loss? The study focuses on the following objectives:

- i) investigate the impact of COVID-19 on learning loss, focusing on the hypothesis that specific demographics of students experienced more significant losses than others.
- ii) identify factors that contributed to learning loss.
- iii) identify outlier countries.

Various analytical methods, including descriptive and predictive analytics, will generate actionable insights during this study. Descriptive analytics will be used to summarise data and examine the relationships between different variables. In contrast, predictive analytics will look at the connections between learning loss and various factors, identifying the factors that contributed the most to learning loss. The study aims to understand the pandemic's effect on global education and learning loss and help inform strategies to strengthen global education.

Literature Review

In their study of COVID-19 student learning loss across 35 studies, Patrinos et al. (2022) discovered an average learning loss of 0.17 standard deviation or around half a year's worth of learning; this shows that learning loss is real and substantial. They classified each study according to its geographical region, length of school closure, student education level, sample size, subject(s) analysed, documented learning impact, and increased inequality. Their result showed evidence of pandemic-related learning losses in all examined regions and education levels. They discovered falling test scores in Uganda, Canada, and the Republic of Korea. Additionally, they discovered that the longer schools were closed, the greater the learning losses.

Engzal et al. (2021), in their study, discovered that Dutch primary school children made little progress during remote learning due to poor face-to-face interaction. Schleicher (2020) also found

that COVID-19-related school closures in the OECD and partner nations caused learning loss, particularly for students without access to digital learning resources.

In addition, Dorn et al. (2020) discovered that the pandemic increased learning loss for all children, particularly students of colour and from low-income backgrounds. Blasko et al. (2022) also discovered significant increases in inequality within and between countries. The pandemic may have worsened learning loss and inequality between countries.

In conclusion, the COVID-19 pandemic caused significant learning loss due to socioeconomic inequalities, technology access, and other factors, with specific demographics of students experiencing more significant learning losses than others. The above studies justify our use of the country name, the decline in test scores due to COVID, weeks schools were closed, GDP per capita, the proportion of private schools in a country, internet penetration rate, school quality, stringency of lockdown index, years of schooling, region code, OECD country, school quality scores minus learning loss and the high-income country as variables for our analysis.

Data Analysis and Results

This study aims to analyse the effect of COVID-19 on learning loss in 41 countries and identify factors that contribute to learning loss. In order to accomplish this, descriptive and predictive analytics were performed; the results are presented below.

Data Wrangling

There were 41 observations and 24 variables, with two missing values in the logweeks column. The dataset was further subset to include only 13 variables required for analysis. Table 1 shows more details on the selected variables.

Table 1. Study's Dataset

Column Name	Data Type	Description
Country	Nominal	Country name
Loss	Numeric	Decline in test scores due to COVID school closures in standard deviations 2021
Weeks	Numeric	Weeks schools closed on average 2021
Gdp	Numeric	GDP per capita 2020
Private	Numeric	Proportion of private schools in country, primary, 2019
Internet	Numeric	Internet penetration rate - individuals using the Internet (% of population)
Hlo	Numeric	School quality
Stringency	Numeric	Stringency of lockdowns index, average 2021
Schooling	Numeric	Years of schooling
Region-code	Nominal	Region code
Oecd	Binary	OECD country
Covid	Numeric	Hlo scores minus learning loss
High	Binary	High income country

Source: Learning Loss COVID-19 2020-2022, The World Bank Data Catalogue (World Bank, 2022)

Key Descriptive Statistics

In this study, key descriptive statistics on learning loss reveal insights into central tendency and dispersion. Table 2 shows a minimum learning loss of 0 and a maximum of 0.8. The average is

0.23, with a median of 0.22, which means 50% of the sample had a learning loss of 0.22 or less. The range (0.8) represents the overall spread, while the interquartile range (0.15) measures the middle 50% spread, less influenced by extreme values. The standard deviation is 0.16, indicating the average deviation of learning loss values from the mean.

Table 2. Key Descriptive Statistics for learning loss

Index	Learning Loss
Minimum	0.00
1 st Quartile	0.14
Median or 2 nd Quartile	0.22
Mean	0.23
3 rd Quartile	0.29
Maximum	0.80
Range	0.80
Interquartile Range	0.15
Variance	0.03
Standard Deviation	0.16

Data Visualisation

Table 3 and Figure 1 shows that the learning loss distribution is right-skewed, indicating more countries in lower intervals. Most students (78%) experienced learning loss between 0 and 0.3 SD, with the highest frequency, 32% occurring in the 0.2-0.3 SD interval.

Table 3. Frequency distribution table of learning loss.

Interval(in S.D)	Frequency	Relative Frequency
$-0.1 < X \leq 0$	1	0.02
$0 < X \leq 0.1$	7	0.17
$0.1 < X \leq 0.2$	11	0.27
$0.2 < X \leq 0.3$	13	0.32
$0.3 < X \leq 0.4$	4	0.10
$0.4 < X \leq 0.5$	3	0.07
$0.5 < X \leq 0.6$	1	0.02
$0.6 < X \leq 0.7$	0	0.00
$0.7 < X \leq 0.8$	1	0.02

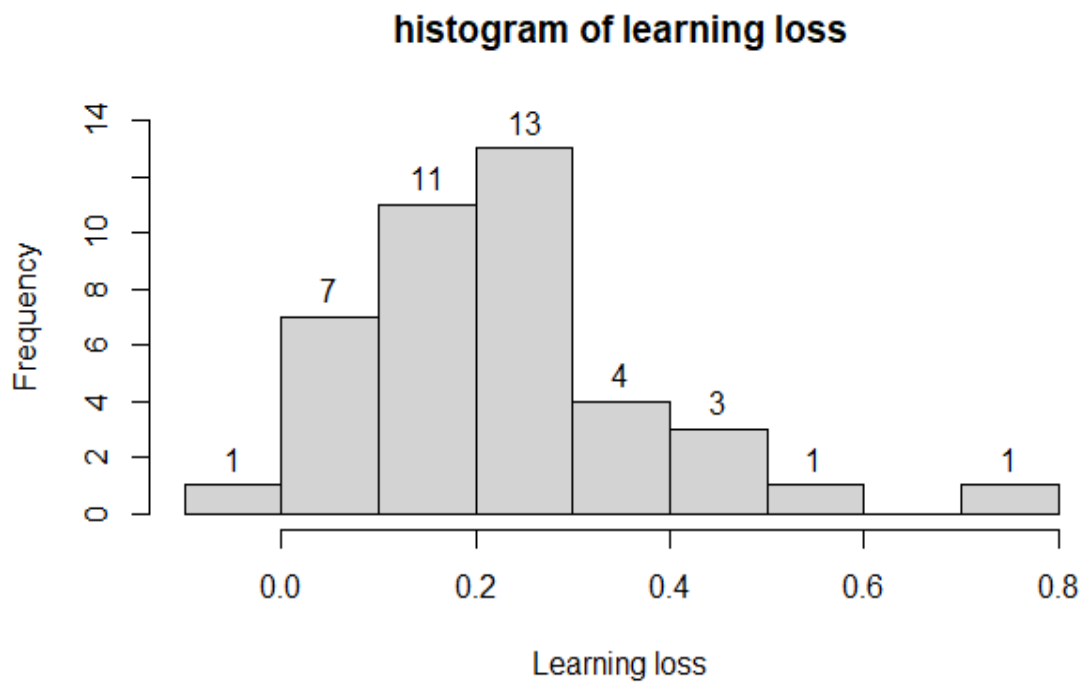
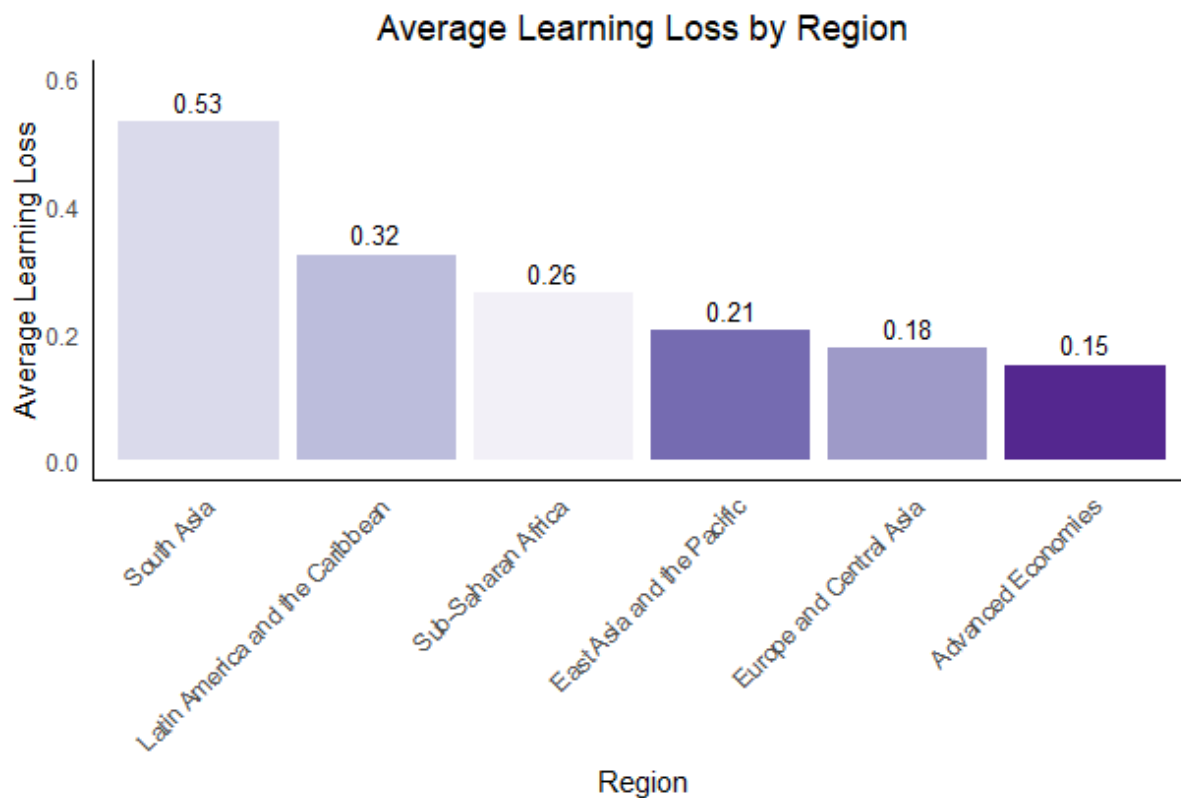


Figure 1. Histogram of Learning loss

What is the average learning loss across regions, income levels, and OECD membership?

Figures 2, 3 and 4 show average learning loss by region, income, and OECD membership, respectively. South Asia had the highest loss (0.53), while Advanced Economies had the lowest (0.15). High-income countries had lower learning loss (0.14) than low-income countries (0.29), reflecting potential disparities due to school closures. This supports Dorn et al.'s (2020) findings on increased learning loss for low-income children. While OECD countries had a lower average learning loss (0.16) than non-OECD countries (0.32), suggesting factors associated with OECD membership, such as wealth or education infrastructure, might contribute to mitigating the adverse effects of school closures.



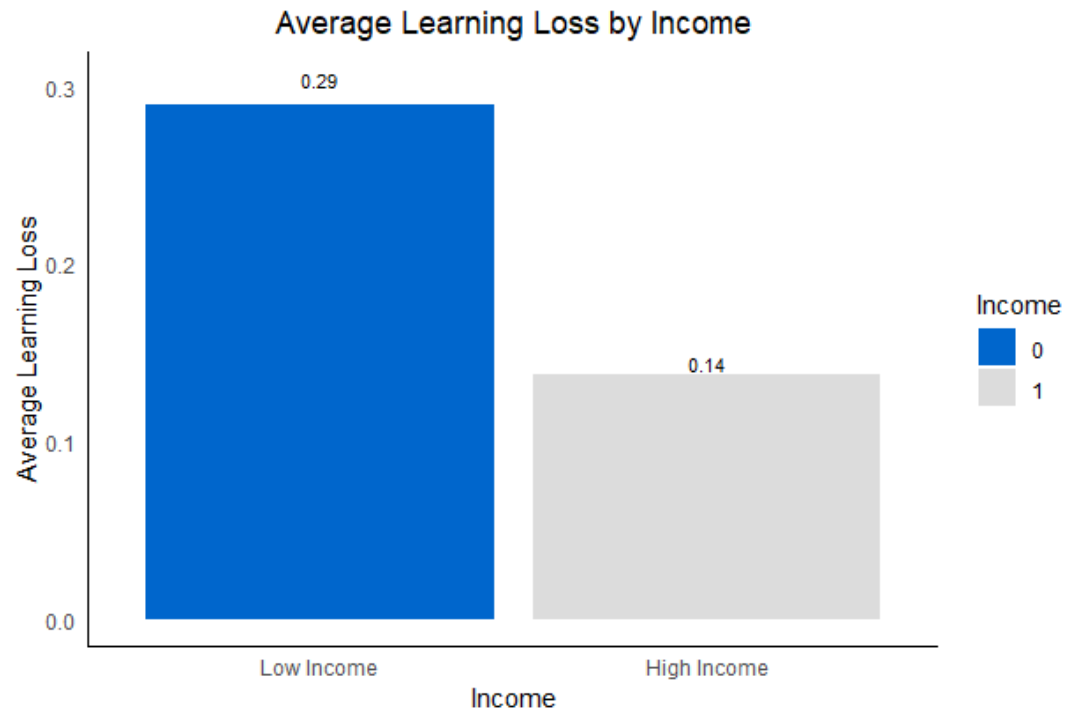


Figure 3. A column chart showing the average learning loss in low-income and high-income countries.

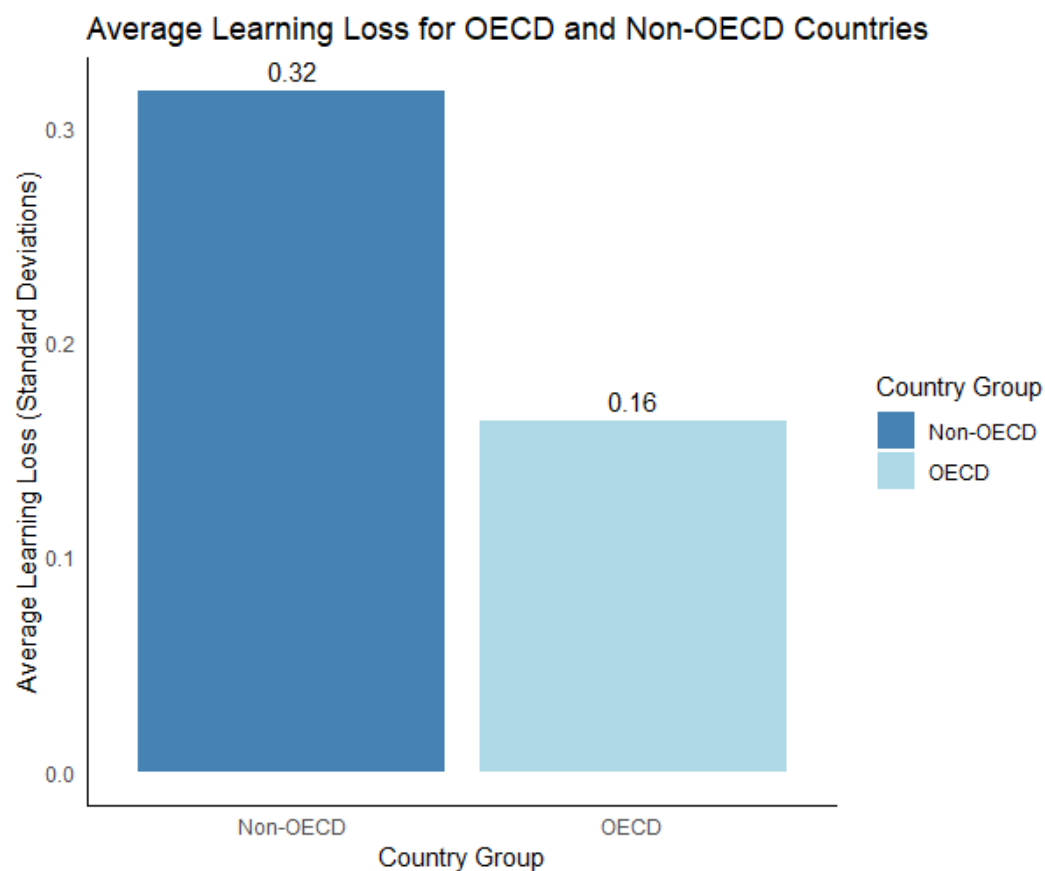


Figure 4. A column chart showing the average learning loss in OECD and non-OECD countries.

Which countries are outliers in terms of learning loss?

Figure 5 shows the box plot of learning loss, which reveals that Nepal (0.8) and India (0.52) are outliers with higher learning losses than most countries. It is essential to retain these outliers because they provide valuable insight into extreme cases.

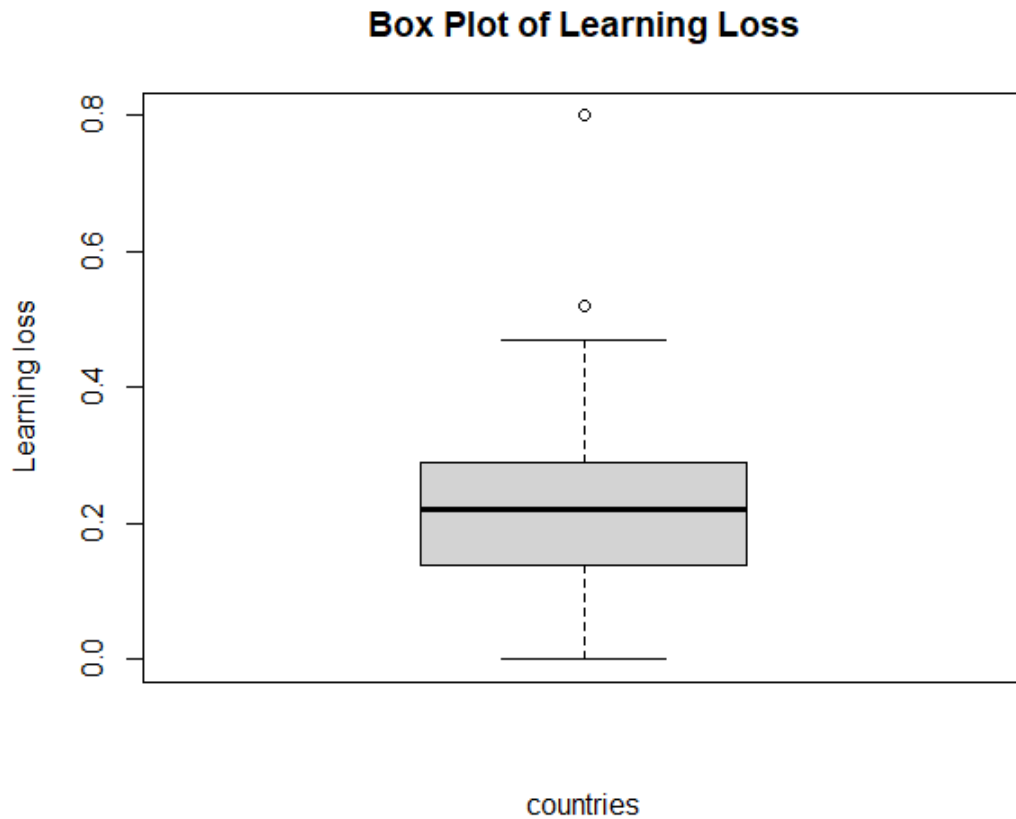


Figure 5. Box plot of learning loss identifying outlier countries

Correlational Analysis

Is there a relationship between learning loss and weeks school closed?

The correlation of 0.64 and Figure 6 demonstrate a strong positive linear relationship between learning loss and weeks of school closure, confirming the findings of Patrinos et al. (2022) that more extended school closures resulted in greater learning losses.

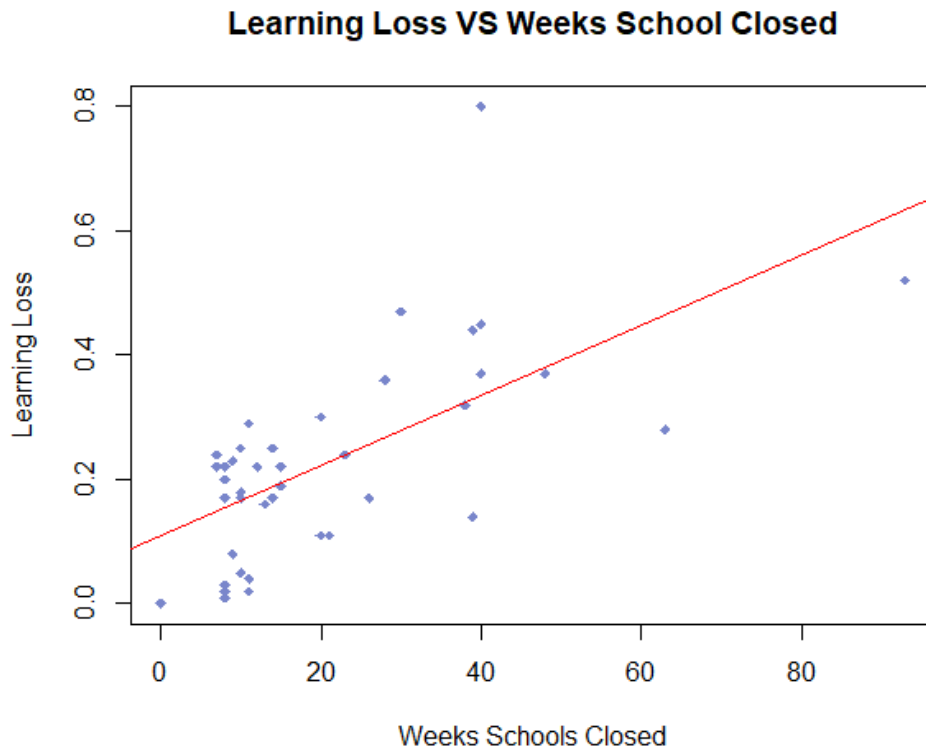


Figure 6. Scatter plot of learning loss and weeks school closed.

What is the correlation between the numerical variables in the study?

Learning loss has a strong positive relationship with school closure weeks (0.64) but negative relationships with COVID (-0.46), GDP (-0.45), internet access (-0.44), and schooling (-0.57), as shown in Figure 7. This suggests that countries with a higher GDP and greater internet access experienced less learning loss, highlighting the significance of educational and economic resources in mitigating the effects of school closures.

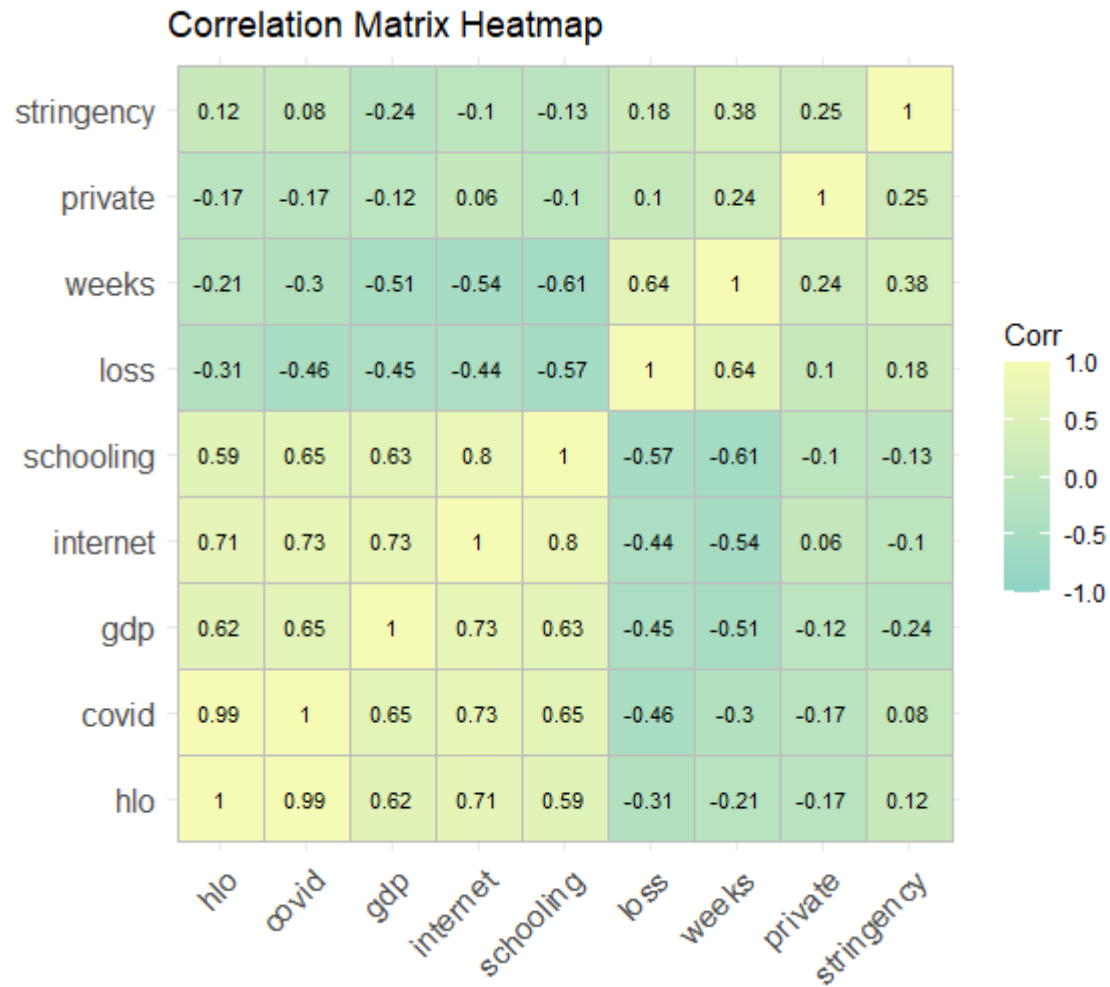


Figure 7. Correlation matrix heatmap of the numerical variables.

Regression

Table 4: Models metrics evaluation

Model	Adjusted R-squared	Residual Standard Error	F-statistic p-value
A	0.5021	0.1197	0.0002275
B	0.5193	0.1177	6.958e-05
C	0.4942	0.1207	5.667e-05

D	0.4924	0.1209	2.045e-05
E	0.4822	0.1221	7.797e-05
F	0.4826	0.1221	2.698e-05

Table 4 shows Model B has the highest adjusted R-square (0.5193) among all models, explaining 52% of the variation in learning loss through weeks, covid, the internet, and schooling. The multiple linear regression equation is given as follows:

$$\text{loss} = 0.46 + (0.0046 * \text{weeks}) + (-0.0006 * \text{covid}) + (0.0024 * \text{internet}) + (-0.0233 * \text{schooling}).$$

The null hypothesis is rejected since there is enough evidence to show that at least one coefficient, weeks, is statistically strong to explain variability in learning loss (p-value = 0.0023). This suggests that the number of weeks of schooling is a significant factor in explaining learning loss.

Assumption tests for Model B show no strong evidence of multicollinearity (all VIF values < 10), a linear relationship between variables, and normally distributed residuals. Figures 8 and 9 present visualisations for linearity and normality. However, the model failed the homoscedasticity assumption (Breusch-Pagan test p-value = 0.01282). While Model D exhibits superior predictive performance for cross-validation, as shown in Table 5, achieving the lowest root mean squared error (0.1048) in cross-validation among all models tested. This indicates better prediction accuracy for Model D.

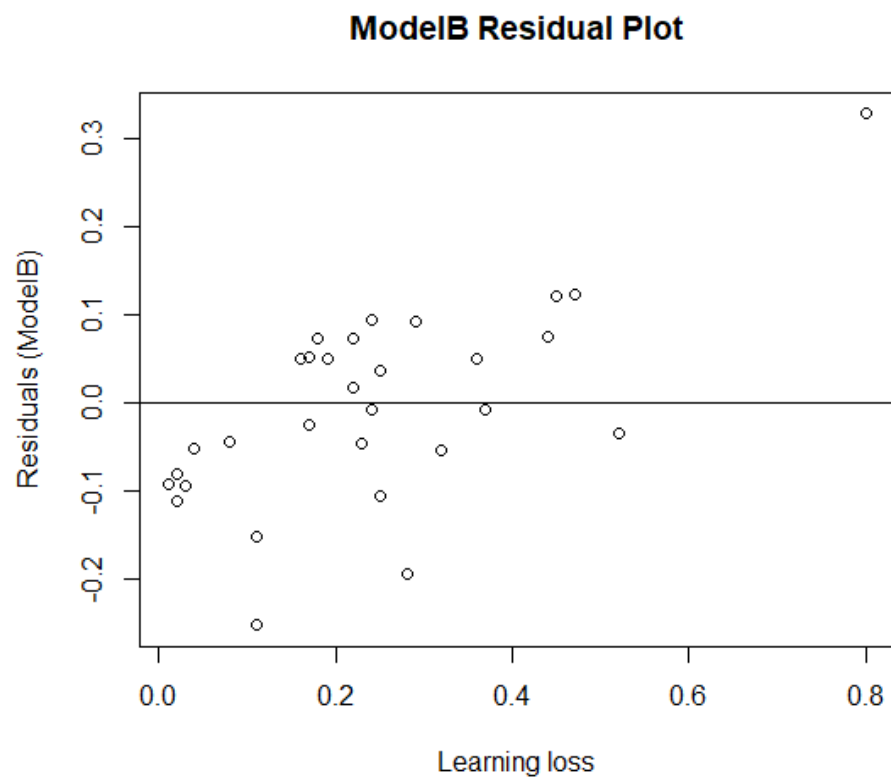


Figure 8. Residual plot for Model B learning loss.

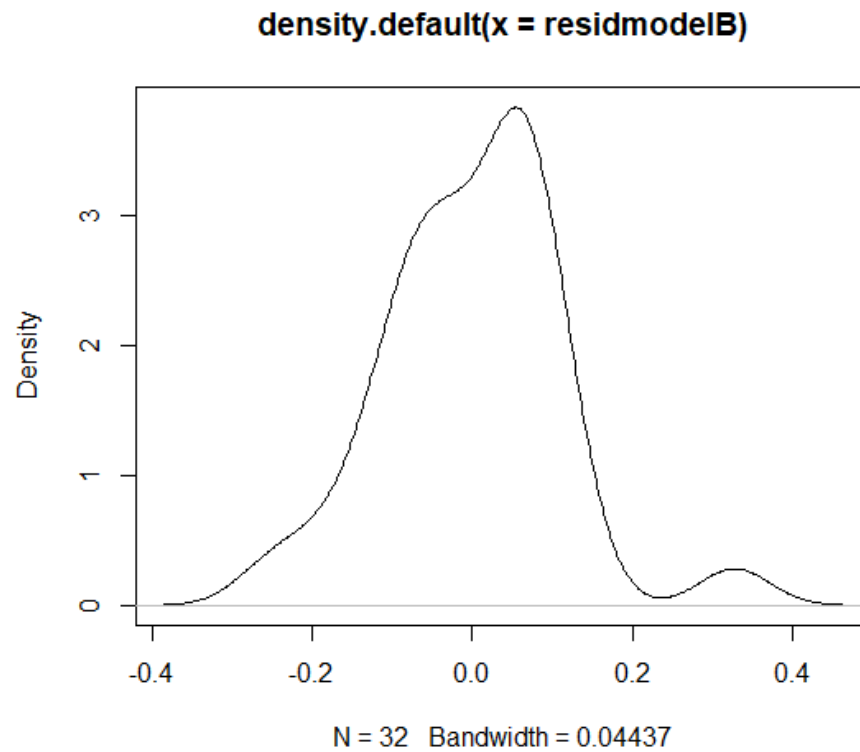


Figure 9. Residuals density plot for Model B to check for normality.

Model	Root mean squared error(RMSE)
A	0.1141
B	0.1136
C	0.1136
D	0.1048
E	0.109
F	0.1134

Table 5. Root mean squared error for all the models.

Conclusion and Limitations.

The pandemic has led to significant learning loss, with specific demographic groups experiencing more severe losses than others. The analysis showed that South Asia, non-OECD, and low-income countries experienced the highest rates of learning loss, corroborating the findings of Dorn et al. (2020), which indicated that the pandemic exacerbated learning loss for all children, particularly those from low-income backgrounds. The strong positive correlation between learning loss and weeks of school closure aligns with the findings of Patrinos et al. (2022). Furthermore, countries with higher GDPs and greater internet access witnessed less learning loss, highlighting the importance of economic and educational resources in mitigating the effects of school closures.

The duration of school closures emerged as a significant predictor of learning loss. This finding affects policymakers, educators, and stakeholders, especially in low-income and non-OECD countries. They can devise strategies to address learning loss and alleviate the negative impacts of school closures, such as investments in education infrastructure, technology, and pandemic management.

However, this study has limitations, including a small sample size and potential issues with data representativeness. Future research could expand its scope by incorporating additional countries to understand the situation comprehensively.

References

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Appendix

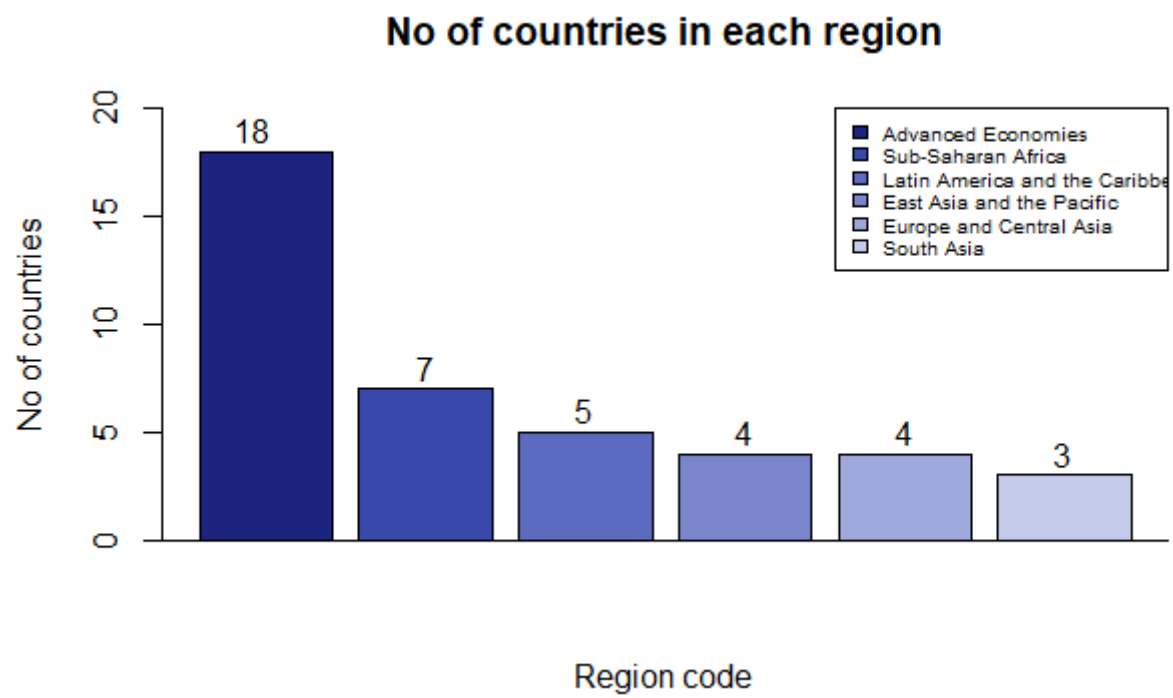


Figure 2b. A column chart showing number of countries in each region.

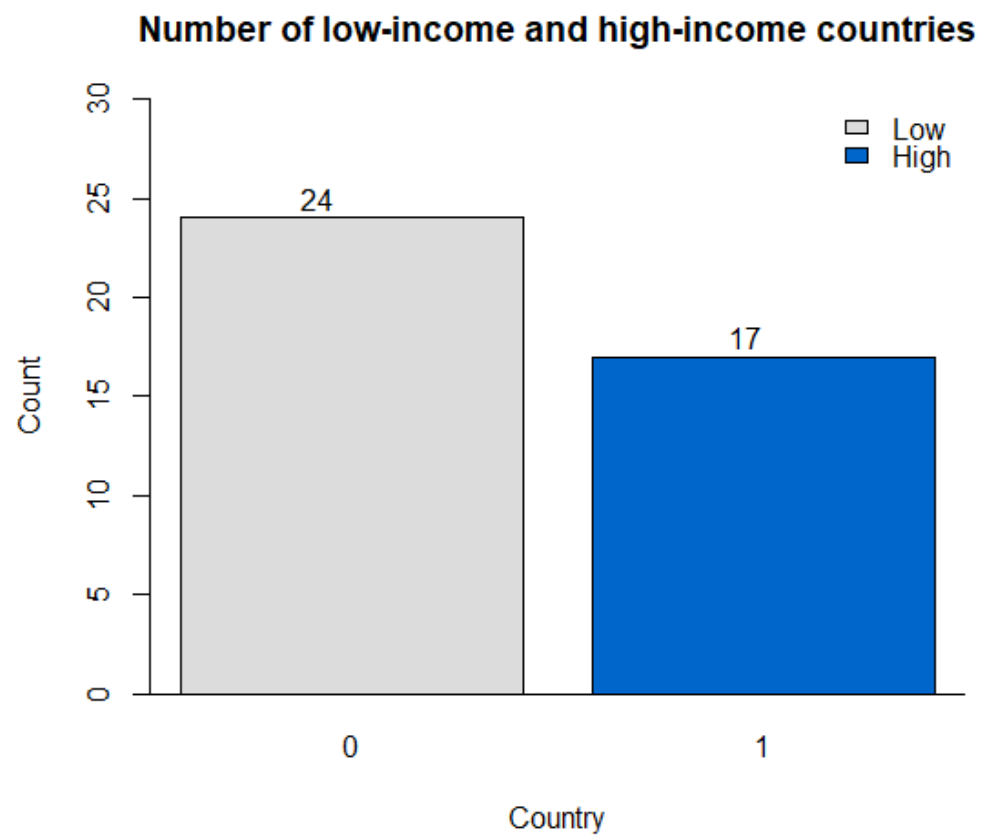


Figure 3b. A column chart showing number of low-income and high-income countries.

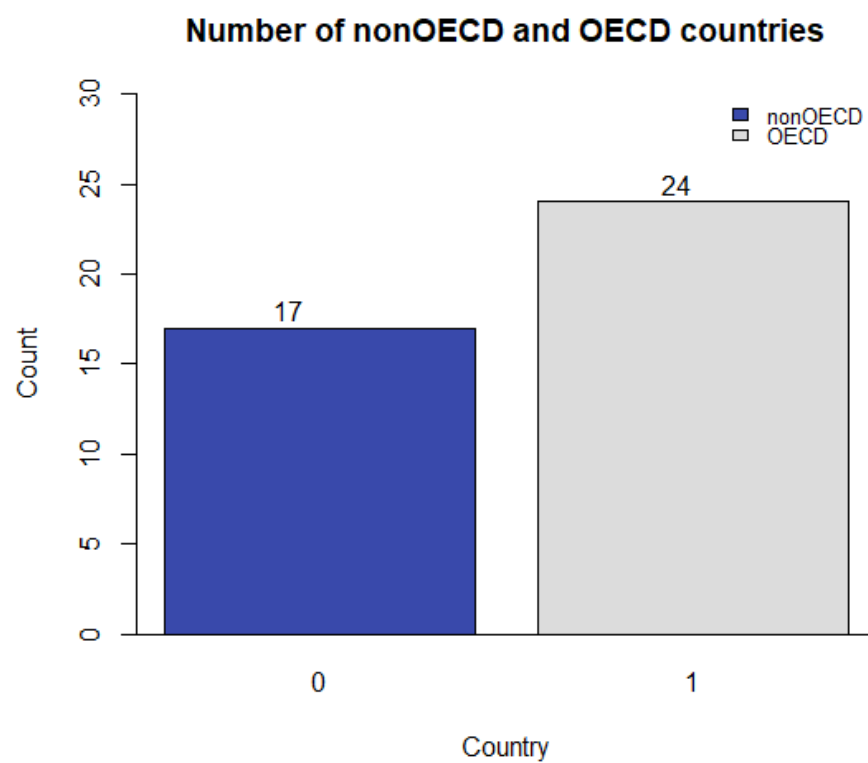


Figure 4b. A column chart showing number of OECD and non OECD countries.